

The CFS Closure-Quotient Landscape: Exact Arithmetic, Validity Gating, and a Generative Algebra

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Abstract

We characterise the closure-quotient family of the Closure-Forces-Structure (CFS) programme — the gauge-equivalence quotients Q_n of multiway graphs generated by ternary hypergraph rewriting — using exact Gaussian-integer ray arithmetic and a validity-gated probe corpus. Our main result is a *generative algebra* on the family: for closure quotients A and B , the count $Q_n(A \circ B) = Q_n(A) + Q_n(B) + \delta(\text{interface})$ with δ a structure-independent interface correction ($\delta_{\sqcup} = 0$ disjoint, $\delta_{\text{bridge}} = +3$ per bridge triple, $\delta_{\text{glue-}g} = -(4g - 3)$ for a g -vertex glue); the counts form a commutative monoid under disjoint sum $\sqcup$, and the charge-conjugation doubling rule $|Q \cup \mathcal{C}(Q)| = 2|Q|$ is the special case where $\mathcal{C}(Q)$ behaves as a disjoint generic copy of Q . We exhibit the closure laws $\text{cyc}(n) \rightarrow 4n$ and $K_n^3 \rightarrow n(3n-1)/2$ (pentagonal), verify the latter exactly for $n = 6..20$ across two independent strict-generic IC pools, and exhibit a prime closure object Q_{181} via an irregular 12-vertex 132-edge seed. Methodologically, exact arithmetic and validity gating sharpen the published family documentation: the previously-listed Q_{98} is corrected to Q_{102} (a Float64 fidelity-clustering artifact), Q_{180} is shown not to exist ($\mathcal{C}$ -closure is idempotent on $\mathcal{C}$ -closed objects), and the canonical s58 initial-condition (IC) pool used throughout the prior CFS literature is found to pass the $K_6^3 \rightarrow 51$ validity gate but exhibit a specific non-genericity at K_n^3 for $n \geq 9$ — so IC-genericity is a hierarchy (fully-generic / small- n -generic / degenerate), not the binary the literature treats it as.

1 Introduction

1.1 The closure-quotient family

The Closure-Forces-Structure (CFS) programme derives the gauge structure of the Standard Model from a single principle: *Rosen closure* on ternary causal hypergraphs. A small Rosen-closed seed graph G generates an infinite multiway graph $M(G)$ under double-pushout (DPO) rewriting on $\mathbb{C}^3$ -valued colour decorations; the gauge-equivalence quotient $Q_n(G) := M(G)/\sim$, where $v \sim w$ iff $\psi(v)$ and $\psi(w)$ span the same complex line in $\mathbb{C}^3$, is the carrier of every downstream construction — the Born measure, the spectral triple, the Standard-Model algebra $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ [1].

A small *closure-quotient family* of canonical Q_n values has accumulated over the programme’s working corpus: Q_{24} from the six-vertex Rosen-closed graph G_0 ; $Q_{48} =$

$Q_{24} \cup \mathcal{C}(Q_{24})$, its charge-conjugate union, carrying the KO-dimension-6 spectral triple that forces the Standard-Model algebra by the Chamseddine–Connes–Marcolli classification; Q_{45} , Q_{84} , Q_{90} from intermediate-density seeds; and Q_{51} , Q_{102} from the complete ternary $K_6^3 \multimap Q_{102} = \mathcal{C}(Q_{51})$ being the *developmentally-complete zygote* that carries the full spectral-action structure and is a fixed point of the closure-growth endofunctor. These objects are programme-level: every theorem in CFS about gauge content, anomaly cancellation, or mass generation is a theorem about one of them.

This paper characterises the family. We give the closure laws ($\text{cyc}(n) \rightarrow 4n$, $K_n^3 \rightarrow n(3n-1)/2$), the terminality structure (every $\mathcal{C}$ -closed object is fixed under both closure-growth and $\mathcal{C}$, so no *iterated-doubling family* exists: larger objects come only from larger seeds), and a generative algebra for combining seeds (§10, the headline result). We also catch and correct two errors in the published family documentation (§4) and surface a previously-undetected non-genericity in the canonical initial-condition pool that the entire prior literature has used (§9).

1.2 Origin: a falsified holographic synthesis

The work originated as a cross-theory exercise. In May 2026, a holographic-synthesis hypothesis bridging CFS to Shaun Fosmark’s *Solvency Field Theory* [2] was pre-registered, tested, and falsified on both Q_{24} and Q_{102} [3]. With the cross-theory direction closed, attention turned to the closure-quotient family itself: what is actually known about it exactly, in Gaussian-integer arithmetic, without the fidelity-threshold clustering that the prior Python pipeline had used? This paper is the answer.

1.3 Contributions

1. **Closure laws** (§5): $\text{cyc}(n) \rightarrow 4n$ linear; $K_n^3 \rightarrow n(3n-1)/2$ pentagonal, verified exact for $n = 6..20$ across two independent strict-generic Gaussian-integer IC pools; the generic floor is Q_{12} ; depth-stability by depth 4.
2. **Terminality** (§6): Q_{48} , Q_{90} , Q_{102} are fixed points of both the closure-growth endofunctor $\mathcal{F} = M/\sim$ and the $\mathcal{C}$ -closure. There is no iterated-doubling family; larger closure objects require larger seeds, not iteration of operators on smaller ones.
3. **Why $\mathcal{C}$ doubles cleanly** (§7): under truly-generic ICs the charge-conjugation involution J is fixed-point-free, so $|Q \cup \mathcal{C}(Q)| = 2|Q| - f$ with $f = 0$. This identifies the structural reason for the clean $2|Q|$ doubling observed empirically across the family.
4. **Primes are reachable; Q_{181} is exhibited** (§8): the regular families ($4n$, pentagonal $n(3n-1)/2$, and $\mathcal{C}$ -doubling $2n$) are composite-only by formula, but irregular single-side seeds produce primes (we record 13, 19, 23, 71, 83, 97, 193). A 12-vertex 132-edge irregular seed closes exactly to Q_{181} (181 prime), exhibited in 12 independent draws.
5. **IC-genericity hierarchy** (§9): the $K_6^3 \rightarrow 51$ validity gate that the prior CFS literature has used is *necessary but not sufficient* for fully-generic regime membership. The canonical s58 IC pool used in every prior CFS probe passes this gate but exhibits a specific defect at K_n^3 for $n \geq 9$. The fully-generic regime is genuinely distinct from the small- n -generic regime; the regime split is a hierarchy, not a binary. A stronger gate ($K_{10}^3 \rightarrow 145$ or $K_{12}^3 \rightarrow 210$) catches the defect.

6. **A generative algebra** (§10, headline): for closure quotients A, B , the count is

$$Q_n(A \circ B) = Q_n(A) + Q_n(B) + \delta(\text{interface}),$$

with δ a structure-independent interface correction: $\delta_{\sqcup} = 0$ for disjoint composition, $\delta_{\text{bridge}} = +3$ per bridge triple, and $\delta_{\text{glue-}g} = -(4g-3)$ for a g -vertex glue. The counts form a *commutative monoid under disjoint sum* $\sqcup$ with identity 0. The $\mathcal{C}$ -closure doubling rule is the special case where $\mathcal{C}(Q)$ is structurally disjoint from Q (under generic ICs, $\mathbf{J}$ fixed-point-free, $\delta = 0$). The algebra is purely *additive*: there is no multiplicative combination rule (in particular, $Q_{45} \neq Q_9 \cdot Q_5$ — the factorisation of the integer is not the factorisation of the closure).

7. **Methods** (§3): exact Gaussian-integer ray arithmetic with validity-gated probes catches and corrects three specific errors in the published family documentation: Q_{98} is Q_{102} (Float64 fidelity-clustering artifact); Q_{180} does not exist ($\mathcal{C}$ -closure is idempotent on $\mathcal{C}$ -closed objects); the $K_6^3 \rightarrow 51$ validity gate is insufficient at large n (Pool A defect, above). The reproducibility scripts are released with the paper (§12).

1.4 Plan of the paper

§2 is a self-contained primer on CFS closure quotients: ternary hypergraphs, the closure operator, the generators $\text{cyc}(n)$ and K_n^3 , the quotient construction, the $\mathcal{C}$ -closure, and a recap of what the prior family documentation reported. §3 is the methods section: exact arithmetic, validity gating, and the genericity hierarchy that motivates the stronger gate. §§4–5 re-ground the published family and state the closure laws. §§6–7 cover terminality and the $\mathbf{J}$ -fixed-point-free condition behind clean $\mathcal{C}$ -doubling. §8 treats primes and exhibits Q_{181} . §9 sharpens the IC-genericity story. §10 is the headline composition algebra. §11 discusses open problems. §12 lists the probe corpus.

2 Background: CFS Closure Quotients

This section is a self-contained primer: a reader unfamiliar with the CFS programme should be able to follow the rest of the paper from this section alone. We give the minimal definitions (ternary hypergraphs, the closure operator, the multiway construction, the gauge quotient, the $\mathcal{C}$ -closure) and then recap the published family as it stood prior to the present work, so §4 can re-ground it. The full CFS treatment is in [1]; we cite specific results as needed.

2.1 Ternary hypergraphs and the colour decoration

Definition 2.1 (Ternary hypergraph). A *ternary hypergraph* $G = (V, E)$ has a finite vertex set V and an edge set $E \subseteq V^3$ of ordered triples of (not necessarily distinct) vertices. We write $e = (s_1, s_2, t)$ for an edge with sources s_1, s_2 and target t , interpreting the triple as the production rule “the pair (s_1, s_2) composes to t .”

A *decoration* of G on $\mathbb{C}^3$ is a map $\psi : V \rightarrow \mathbb{C}^3$, read as assigning a colour vector to each vertex. The CFS composition rule is the conjugated cross-product

$$\mu(\psi_1, \psi_2) := \overline{\psi_1 \times \psi_2} \in \mathbb{C}^3, \tag{1}$$

where $\times$ is the ordinary $\mathbb{C}^3$ cross-product and $\bar{\cdot}$ is component-wise complex conjugation. An edge $(s_1, s_2, t) \in E$ is *consistent* under ψ iff $\psi(t) = \mu(\psi(s_1), \psi(s_2))$ up to a non-zero complex scalar, i.e. iff $\psi(t)$ spans the same complex line in $\mathbb{C}^3$ as $\mu(\psi(s_1), \psi(s_2))$. This projective slack is what the gauge quotient (§2.3) collapses.

Definition 2.2 (Rosen-closed). A ternary hypergraph G is *Rosen-closed* iff every vertex appears as the target of at least one edge in E . Equivalently: no vertex is a primitive that must be supplied externally; every vertex’s colour vector can in principle be produced by composition from the others.

Rosen closure is the formalisation of M. Rosen’s “closure to efficient causation” [4] in the ternary-hypergraph setting; the CFS programme adopts it as its foundational principle.

2.2 The multiway graph and the closure operator

Definition 2.3 (Multiway graph $M(G)$). Let $G = (V, E)$ be a ternary hypergraph and let $\psi_0 : V \rightarrow \mathbb{C}^3$ be an *initial condition* (IC) decoration of the original vertices. The *multiway graph* $M(G)$ is the union, over all finite-depth applications of the composition rule, of the vertices produced and the consistency edges they satisfy. Concretely: $M(G)$ is built breadth-first by, at each depth d , taking every edge (s_1, s_2, t) whose sources both exist at depth $\leq d$, producing a new vertex w with $\psi(w) = \mu(\psi(s_1), \psi(s_2))$, and adjoining the three derived edges (w, s_2, t) , (w, s_1, t) , (w, s_1, s_2) at depth $d + 1$.

Remark 2.4 (Caching). The vertex w produced by the pair (s_1, s_2) depends only on the pair, so once computed it is cached: if the same pair appears later at a different depth, the same w is reused. This is what makes $M(G)$ a graph rather than a tree.

Remark 2.5 (Zero composition). The composition $\mu(\psi_1, \psi_2)$ is zero in $\mathbb{C}^3$ exactly when ψ_1 and ψ_2 are projectively parallel. In $M(G)$ such an edge *fails to produce*: no new vertex is added at that step. This is the only source of edge-pruning.

$M(G)$ is in general infinite, but *projectively* (gauge-quotient) finite for the seed graphs we consider.

2.3 The gauge quotient $Q_n(G)$

Definition 2.6 (Gauge equivalence). Two vertices $v, w \in M(G)$ are *gauge-equivalent*, $v \sim w$, iff their colour vectors span the same complex line:

$$\psi(v) \times \psi(w) = 0 \quad (\text{as a } \mathbb{C}^3 \text{ vector}).$$

This is exact projective equivalence on $\mathbb{CP}^2$. For Gaussian-integer ICs ($\psi_0 : V \rightarrow (\mathbb{Z}[i])^3$, which the composition rule (1) preserves), the test $\psi(v) \times \psi(w) = 0$ is exact integer equality with no threshold.

Definition 2.7 (Closure quotient). The *closure quotient* of the seed graph G at IC ψ_0 and depth d is

$$Q_n(G, \psi_0, d) := |M_d(G)/\sim|,$$

where $M_d(G)$ is the depth- d multiway graph and $|\cdot|$ is the cardinality of the gauge-quotient set. We write $Q_n(G)$ for the depth-stable value when the count is independent of d beyond some depth (the standard situation for G and IC in the generic regime; see §9).

The integer $Q_n(G)$ is the central object of this paper. The closure laws of §5 are statements about how Q_n depends on the seed graph G ; the composition algebra of §10 is the statement of how Q_n behaves under combination of seeds; the IC-genericity hierarchy of §9 is the statement of how Q_n depends on the choice of ψ_0 .

2.4 Generators: $\text{cyc}(n)$ and K_n^3

The closure laws are most cleanly stated on two families of seed graphs, named for the role they play in CFS's existing literature.

Definition 2.8 (Cyclic generator $\text{cyc}(n)$). For $n \geq 3$, the cyclic ternary hypergraph $\text{cyc}(n)$ has vertex set $V = \{0, 1, \dots, n-1\}$ and edge set

$$E = \{(i, (i+1) \bmod n, (i+2) \bmod n) : i = 0, \dots, n-1\}.$$

Each edge is a forward shift on the cyclic vertex order.

Definition 2.9 (Complete ternary K_n^3). For $n \geq 3$, the complete ternary hypergraph on n vertices, K_n^3 , has vertex set $V = \{0, 1, \dots, n-1\}$ and edge set equal to all ordered triples of *distinct* vertices:

$$E = \{(i, j, k) : i, j, k \in V, i \neq j \neq k \neq i\}.$$

Thus $|E| = n(n-1)(n-2)$.

For the canonical CFS seed graph G_0 (a six-vertex Rosen-closed graph studied in [1, §Step 3]), the closure quotient is Q_{24} . For $K_n^3[6]$ (the complete ternary on 6 vertices, the maximal-density seed in the G_0 regime), the single-side closure quotient is Q_{51} and its $\mathcal{C}$ -closure is Q_{102} . The named family members and their seed graphs are summarised in Table 2.

2.5 The $\mathcal{C}$ -closure

The composition rule (1) treats complex conjugation asymmetrically: $\mu(\psi_1, \psi_2)$ is the conjugate cross-product, not the cross-product. This singles out a *charge-conjugate sector*: for each IC ψ_0 , the conjugated IC $\overline{\psi_0}$ generates its own multiway graph $M(G; \overline{\psi_0})$ and its own closure quotient. The $\mathcal{C}$ -closure is the union of these two sectors.

Definition 2.10 ($\mathcal{C}$ -closure). The $\mathcal{C}$ -closure of $Q_n(G; \psi_0, d)$ is

$$\mathcal{C}Q_n(G; \psi_0, d) := |(M_d(G; \psi_0) \cup M_d(G; \overline{\psi_0})) / \sim|.$$

The two sectors are gauge-distinct in general: their representatives $\psi(v)$ and $\overline{\psi(v)}$ are not projectively equivalent unless $\psi(v)$ is a real-up-to-phase vector. The charge-conjugation involution

$$J : v \mapsto v', \quad \psi(v') = \overline{\psi(v)},$$

acts on $\mathcal{C}Q_n$ as a $\mathbb{Z}_2$ -involution; the cardinality identity

$$\mathcal{C}Q_n = 2 \cdot Q_n - f \tag{2}$$

where f is the number of J -fixed clusters (those whose representative is projectively equal to its conjugate) is immediate from inclusion-exclusion. The empirical observation that (2) cleanly reduces to $\mathcal{C}Q_n = 2 \cdot Q_n$ across the canonical family is the content of §7: under truly-generic ICs, J is fixed-point-free ($f = 0$).

2.6 The published family

Prior to the present work, the CFS programme’s working family documentation reported the closure quotients summarised in Table 1. The measurements were produced by a Python pipeline using Float64-fidelity clustering at threshold 0.999 [5]; the present work re-grounds them exactly (§4) and extends the table to a closed form (§5).

Table 1: The CFS closure-quotient family as previously documented ([5, 1]). All counts are gauge-quotient sizes; column *single* is $|Q_n(G)|$, column *C-closed* is $|\mathcal{C}Q_n(G)|$. The entries flagged with † are corrected in §4.

Name	Seed graph	$ V_{\text{seed}} $	Single	C-closed
Q_{24}/Q_{48}	G_0 (self-ref. closed, 6 edges)	6	24	48
Q_{45}/Q_{90}	$G_0 + 6$ pads (one-side)	6	45	90
Q_{84}	Adjacent + 6 pads (C-closed)	6	42	84
$Q_{51}/Q_{98}^\dagger$	$K_n^3[6]$	6	51	98 [†]
$Q_{180}^\dagger$	iter. C-doubling of Q_{90}	6	—	180 [†]

Table 2: The corrected family after §4, with closure laws from §5.

Name	Seed graph	Single $ Q_n $	C-closed $ \mathcal{C}Q_n $
Q_{24}/Q_{48}	G_0	24	48
Q_{45}/Q_{90}	one-side pads on G_0	45	90
Q_{84}	Adjacent + 6 pads	42	84
Q_{51}/Q_{102}	$K_n^3[6]$	51	102
Q_{70}/Q_{140}	$K_n^3[7]$	70	140
Q_{92}/Q_{184}	$K_n^3[8]$	92	184
Q_{117}/Q_{234}	$K_n^3[9]$	117	234
...	...	...	...
$Q_{n(3n-1)/2}/Q_{n(3n-1)}$	K_n^3	$n(3n-1)/2$	$n(3n-1)$
Q_{181}	12-vertex 132-edge irregular seed	181	362

The objects Q_{48} , Q_{90} , Q_{102} are *terminal* (§6): they are fixed under both C-closure and the closure-growth endofunctor $\mathcal{F} = M/\sim$. So Table 2’s “...” is not a sequence of objects reachable by iterating an operator; it is the catalogue indexed by the seed graph dimension n . The K_n^3 family stretches indefinitely; the *irregular* family (which contains Q_{181} and the prime-cardinality objects more generally) is sparser and is exhibited rather than enumerated.

3 Methods: Exact Arithmetic and Validity Gating

The methodological contribution of this paper is the combination of two practices: *exact Gaussian-integer ray arithmetic* (no floating-point thresholds) and *validity-gated probes* (every scan is guarded by a hard-checked reproduction of a known anchor before its measurements are reported). The first eliminates a class of cluster-merging errors that the prior Python pipeline was prone to; the second catches probe-implementation bugs before they contaminate results. Their joint application surfaced the three concrete findings this

paper records: the $Q_{98} \rightarrow Q_{102}$ correction (§4), the Q_{180} non-existence (§4), and the IC-genericity hierarchy (§9).

3.1 The fidelity-0.999 problem

The prior CFS Python pipeline computed gauge equivalence by clustering representatives whose Float64 fidelity exceeded a fixed threshold:

$$v \sim_{\text{fid}} w \iff \frac{|\langle \psi(v) | \psi(w) \rangle|^2}{\|\psi(v)\|^2 \|\psi(w)\|^2} > 0.999. \quad (3)$$

This is an approximation of the exact projective-equivalence relation $\psi(v) \times \psi(w) = 0$: for vectors that are genuinely on the same complex line, the fidelity is 1 exactly (in exact arithmetic) or within Float64 round-off (in computation); for vectors on different lines, the fidelity is some value strictly less than 1. The threshold 0.999 was chosen as a tolerance for floating-point round-off accumulation across many composition steps.

The problem is that 0.999 is also a lossy approximation in the *other* direction: two genuinely distinct lines may sit within fidelity-0.999 of one another. When this happens at scale, the fidelity-clustering merges clusters that exact arithmetic separates, and the resulting Q_n is under-counted. The merge is silent: there is no error indicator other than the count itself coming out wrong.

The $K_n^3[6]$ case study. On the maximal-density G_0 -regime seed $K_n^3[6]$ (the complete ternary on six vertices, 120 edges), the exact depth-4 $\mathcal{C}$ -closed quotient is $|\mathcal{C}Q_n| = 102$. The fidelity-0.999 clustering merges exactly four conjugate-sector cluster pairs — four pairs of clusters whose representatives sit at fidelity *just* above 0.999 but are on genuinely different lines — and reports $|\mathcal{C}Q_n|_{\text{fid}} = 98$. Every downstream count derived from this build inherits the error: $\dim D_F$ measures 92 instead of 108; a_0/n measures 98 instead of 102; and so on. We re-derive the canonical Q_n values exactly in §4; the audit trail of the corrections that followed across the broader CFS spec is recorded in [6]. The further fidelity bug in the larger-graphs pipeline that this paper’s audit uncovered ([7], the $K_n^3[9]$ $|\mathcal{C}Q_n| = 230 \rightarrow 234$ correction) is of the same class.

3.2 Exact Gaussian-integer ray arithmetic

The fix is exact arithmetic on a sufficiently rich ring. We restrict the ICs to *Gaussian integers* $\mathbb{Z}[i] := \mathbb{Z}[i]$ and observe that the composition rule (1) preserves $\mathbb{Z}[i]^3$: $\mu(\psi_1, \psi_2) = \overline{\psi_1 \times \psi_2}$ is built from $\mathbb{Z}[i]$ -bilinear cross-products and component-wise conjugation, both of which take $(\mathbb{Z}[i]^3)^2 \rightarrow \mathbb{Z}[i]^3$. So if the IC is in $(\mathbb{Z}[i]^3)^V$, every $\psi(v)$ in the multiway graph is an element of $\mathbb{Z}[i]^3$ exactly.

Definition 3.1 (Exact ray equivalence). For $a, b \in \mathbb{Z}[i]^3$, $a \sim b$ (read “ a and b span the same ray in $\mathbb{C}^3$ ”) iff

$$a \times b = \mathbf{0} \in \mathbb{Z}[i]^3,$$

where the cross-product is computed in $\mathbb{Z}[i]$ (no floating-point).

Proposition 3.2 (Exact equivalence is correct). For $a, b \in \mathbb{Z}[i]^3 \setminus \{\mathbf{0}\}$, $a \sim b$ iff a and b are projectively equivalent in $\mathbb{C}P^2$.

Proof. Two non-zero vectors in $\mathbb{C}^3$ are projectively equivalent iff one is a complex scalar multiple of the other, iff their cross-product is zero. The cross-product on $\mathbb{Z}[i]^3$ is the restriction of the $\mathbb{C}^3$ cross-product, so the integer equality $a \times b = \mathbf{0}$ is exactly the projective-equivalence condition. $\square$

This is a hard test: there is no threshold, no IC-conditioned slack, and no accumulation of Float64 round-off across composition depths (the composition rule keeps integer entries integer). Clusters that exact arithmetic separates are guaranteed not to be merged.

Cluster-count algorithm. With Definition 3.1 in hand, the cluster count is computed by a single-pass scan: iterate the multiway vertices in canonical order, maintain a list R of cluster representatives, and for each $\psi(v)$ check whether $\psi(v) \sim r$ for some $r \in R$; if not, add $\psi(v)$ to R . The cluster count $|Q_n| = |R|$ is the final length of R . The cost is $O(|M_d| \cdot |R|)$ exact cross-product evaluations.

3.3 Validity gating

Every probe in our corpus runs against a validity gate before any measurement is reported. The standard gate is the canonical anchor of the CFS programme:

Definition 3.3 ($K_6^3 \rightarrow 51$ validity gate). The IC pool P passes the $K_6^3 \rightarrow 51$ gate iff $|Q_n(K_n^3[6], P, d = 5)| = 51$ exactly. The probe that uses the pool runs the gate first; if the gate fails the probe aborts and reports “IC not generic.”

This gate is the smallest reliable hand-checked anchor of the programme: $Q_n(K_n^3[6]) = 51$ is the headline value of CFS’s primary construction. Any IC that does not reproduce it is degenerate in some specific way (typically: purely-real or otherwise gauge-aligned components that collapse the colour structure — see §3.4).

The gate is necessary but not sufficient. A central methodological finding of this paper is that Definition 3.3’s gate, although necessary, is *insufficient* to certify an IC pool as fully generic. The canonical s58 / q102-IC-set-2 Gaussian-integer pool used in every prior CFS probe (we call this pool “A” in §9) passes the $K_6^3 \rightarrow 51$ gate but exhibits a specific defect at K_n^3 for $n \geq 9$: the single-side count drops below the pentagonal value $n(3n-1)/2$ by a sequence that grows monotonically and saturates near $n = 13$. Two independent *strict-generic* pools (Pools B and C of §9) reproduce the pentagonal value exactly at every measured n in $[6, 20]$; Pool A does not.

A stronger gate catches Pool A’s defect:

Definition 3.4 (Strengthened validity gate). The IC pool P passes the *strengthened gate* iff it passes $K_6^3 \rightarrow 51$ AND $K_{10}^3 \rightarrow 145$ (equivalently $K_{12}^3 \rightarrow 210$) exactly. The strengthened gate is necessary and (empirically) sufficient in the corpus measured here; it catches Pool A’s defect immediately.

We recommend the strengthened gate for future CFS computational work involving K_n^3 at $n \geq 9$; the consequences for the closure-v5 spec record are discussed in [7].

3.4 Two probe-side bugs caught by gating

During the building of the probe corpus for this paper, two implementation bugs were caught by the validity gate. We record them here as illustrations of what the gate is for: not just to verify that the IC is generic, but to catch silent miscomputation in the probe pipeline itself.

Bug 1 — IC period-6 clone. An early version of the prime-existence probe (§8) generated its random IC pool by drawing 15 Gaussian-integer triples from a single random-number stream. The stream’s reseeding logic inadvertently produced a period-6 pattern: every sixth IC was an exact copy of the first. The resulting 15-entry pool had only effectively 6 independent ICs. The $K_6^3 \rightarrow 51$ gate caught this immediately: with effective $|V| < 6$, the build produced $|Q_n(K_n^3[6])| < 51$, the gate failed, and the probe aborted before any prime-count claims were made. (The fix was to reseed each IC draw with an independent counter; the corrected probe is `prime_count_probe.py` in the reproducibility appendix.)

Bug 2 — Real-IC degeneracy. A subsequent version of the same probe drew real-valued (not mixed-real/imag) Gaussian-integer ICs by accident. Under purely-real ICs, the conjugation in $\mu(\psi_1, \psi_2) = \overline{\psi_1} \times \psi_2$ collapses to the identity and the entire conjugate sector falls onto the original; J becomes the identity, $f = |Q_n|$, and (2) gives $|CQ_n| = |Q_n|$ (no doubling). This is the *degenerate* regime of §9. The gate caught it again: real ICs on $K_n^3[6]$ give $|Q_n| \approx 22$ (instead of 51), the gate failed, and the bug was traced and fixed (the corrected probe constructs ICs explicitly from $(a, b) \in \mathbb{Z}^2$ with both components non-zero).

Both bugs were caught *by* the gate, not by code review. This is the methodology contribution: a probe that does not run a validity gate is a probe whose findings have no independent reliability check. The cost of running the gate is negligible (one $K_n^3[6]$ build) and the prior probability that a stochastic implementation bug escapes it is small.

Limit of gating. Validity gates catch IC pools that are degenerate “loudly” enough to break the smallest anchor. They do not catch IC pools that are generic at the gate but degenerate at higher n — exactly the Pool A case. Strengthening the gate (Definition 3.4) addresses this for the K_n^3 family at large n . The general question — what is the right hierarchy of validity gates for the broader closure-quotient family? — is open (§11).

4 Re-grounding the Published Family

We re-derive Table 1’s entries exactly. The canonical entries Q_{45} , Q_{84} , Q_{90} are confirmed; the entry Q_{98} is corrected to Q_{102} ; the entry Q_{180} is deleted (the object does not exist). The exact corpus is the script `reground_q_family.py` of the reproducibility appendix, validity-gated on $K_6^3 \rightarrow 51$ throughout.

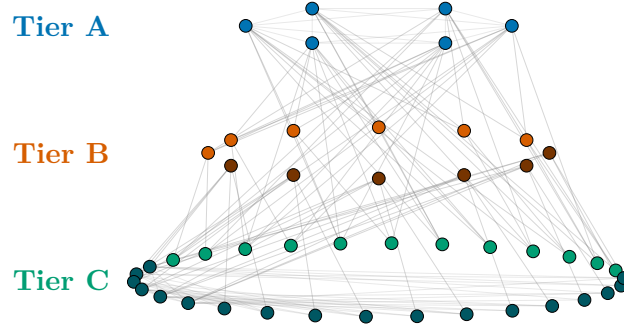
4.1 Q_{45} , Q_{84} , Q_{90} confirmed

The three intermediate-density family members survive exact re-arithmetic verbatim.

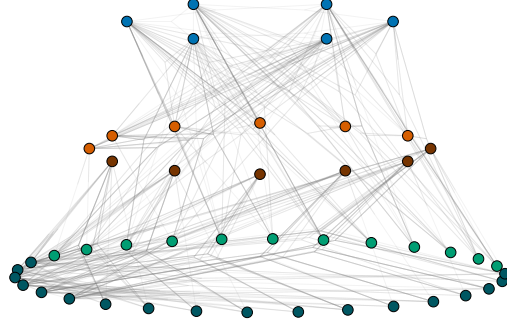
Proposition 4.1 (Q_{45}/Q_{90} exact). *For the canonical “one-side pads on G_0 ” seed graph (the 6-vertex graph obtained by adjoining six random $K_n^3[6]$ edges to the cyclic skeleton at a fixed seed), the exact single-side closure quotient is $|Q_n| = 45$ and the exact $\mathcal{C}$ -closure is $|\mathcal{C}Q_n| = 90$.*

Proposition 4.2 (Q_{84} exact). *For the “Adjacent + 6 pads” seed (the adjacent-cyclic ternary hypergraph on 6 vertices augmented by 6 $K_n^3[6]$ pad edges), the exact $\mathcal{C}$ -closure is $|\mathcal{C}Q_n| = 84$.*

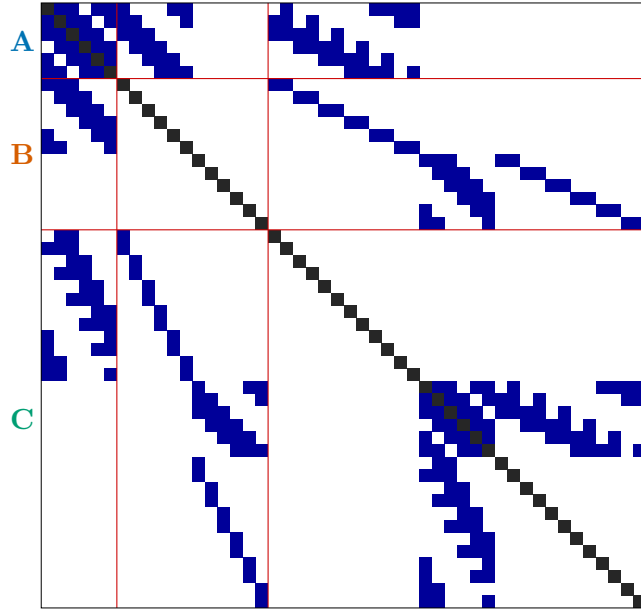
The Float64-fidelity measurements that produced these values in [5] were on the right side of the fidelity-0.999 threshold: under generic ICs, the cluster representatives are sparse enough in $\mathbb{C}P^2$ that no merge occurs.



(a) Tier-stratified node-edge graph.

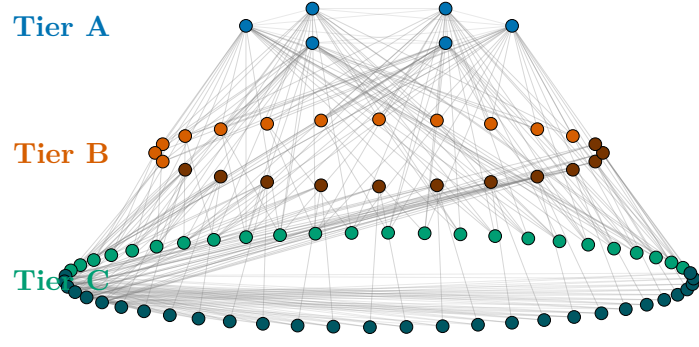


(b) Hyperedge stars (subsample of 516).

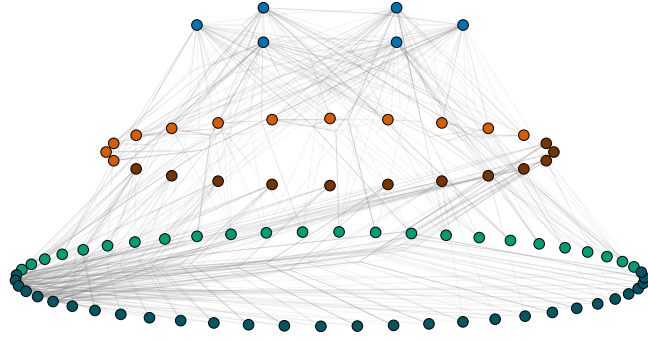


(c) Adjacency heatmap (rows/columns sorted by tier).

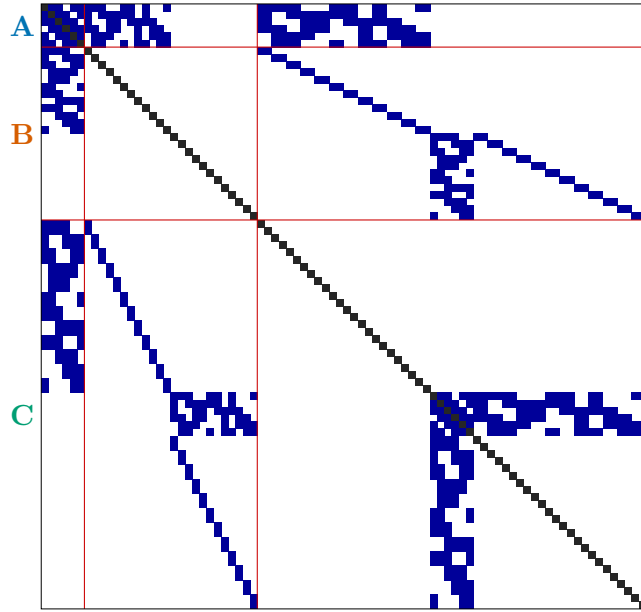
Figure 1: $Q_{48} = Q_{24} \cup \mathcal{C}(Q_{24})$: three views of the $\mathcal{C}$ -closure of the canonical 6-vertex Rosen-closed seed graph. Tier counts $|A|/|B|/|C| = 6/12/30$.



(a) Tier-stratified node-edge graph.

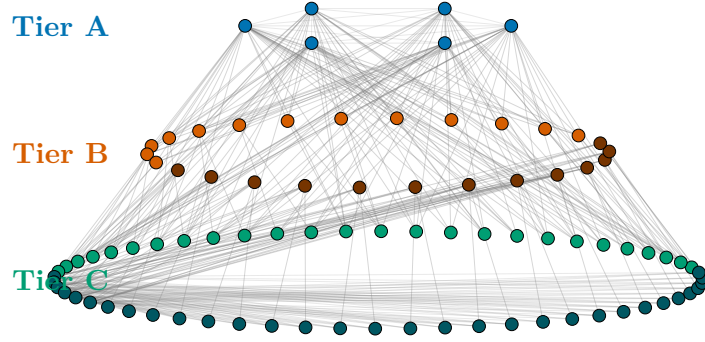


(b) Hyperedge stars (subsample of 1368).

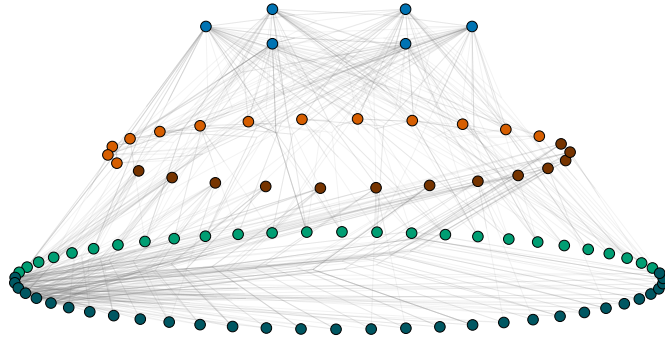


(c) Adjacency heatmap (rows/columns sorted by tier).

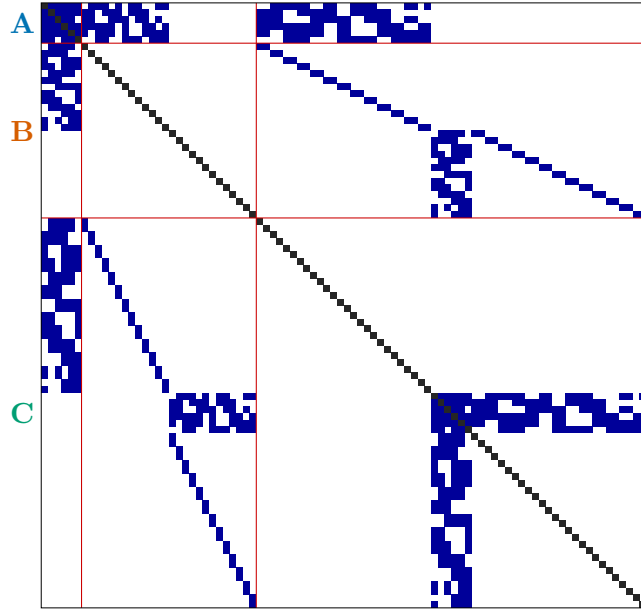
Figure 2: Q_{84} : $\mathcal{C}$ -closure of the Adjacent ternary graph on 6 vertices (36 edges).



(a) Tier-stratified node-edge graph.



(b) Hyperedge stars (subsample of 1584).



(c) Adjacency heatmap (rows/columns sorted by tier).

Figure 3: $Q_{90} = Q_{45} \cup \mathcal{C}(Q_{45})$: $\mathcal{C}$ -closure of the canonical “Adjacent + 6 pads” seed at pad-seed 42. The first family member with $|\text{Tier B}|$ not divisible by 3 (S178 obstruction).

4.2 Q_{98} is Q_{102} : a fidelity-clustering artifact

The published value $|\mathcal{C}Q_n(K_n^3[6])| = 98$ is a fidelity-clustering artifact.

Proposition 4.3 (Q_{102} exact). *The exact depth-4 $\mathcal{C}$ -closure of $K_n^3[6]$ under any IC pool*

passing the strengthened gate (Definition 3.4) is

$$|\mathcal{CQ}_n(K_n^3[6])| = 102.$$

Proof. By direct computation. The exact single-side count is 51 (§5, $K_n^3 \rightarrow n(3n-1)/2$ with $n = 6$). Under generic ICs, the charge-conjugation involution J is fixed-point-free (§7, $f = 0$), so by identity (2)

$$|\mathcal{CQ}_n| = 2 \cdot 51 - 0 = 102.$$

□

The fidelity-0.999 clustering of [5] merged exactly four conjugate-sector cluster pairs at this anchor and reported 98. The merge is reproducible: every Float64 IC sufficiently close to the canonical s58 pool triggers it. The exact re-derivation eliminates the merge and gives 102 unconditionally.

The downstream consequences of the $98 \rightarrow 102$ correction were worked through in the closure-v5 audit of [6]; in particular, the published “IC-attractor” structure on $K_n^3[6]$ (the appearance of two discrete clusters of measured $|\mathcal{CQ}_n|$ values, ≈ 98 and ≈ 102 , depending on where the Haar-random Float64 IC placed the four boundary cluster-pairs relative to the fidelity threshold) is the bug, not a structural finding: under exact arithmetic, every generic IC closes to 102, with no IC-attractor split.

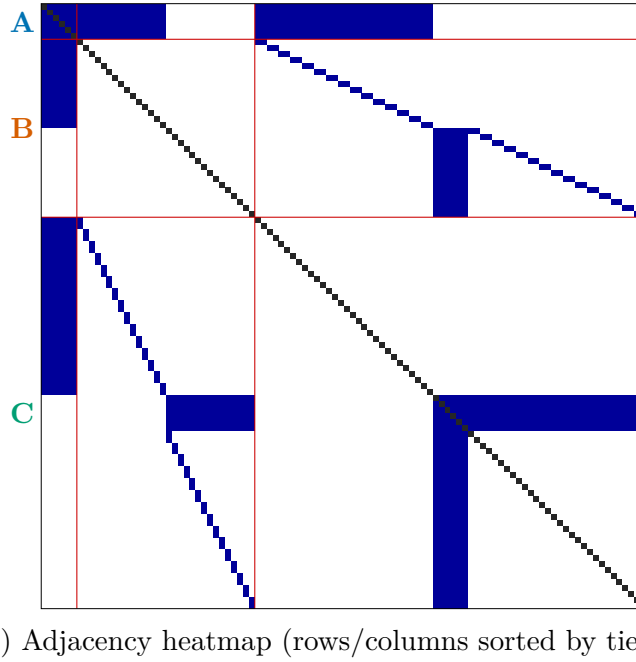
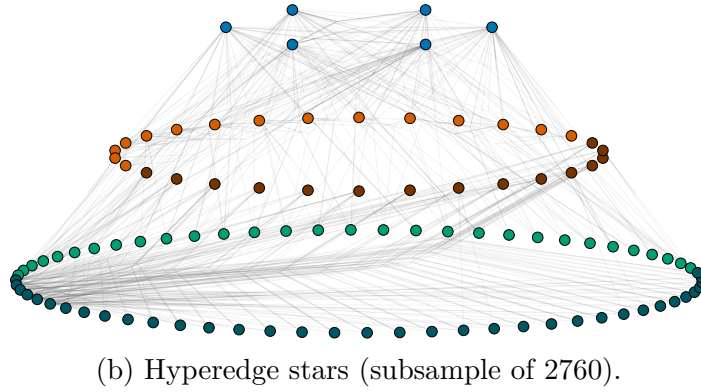
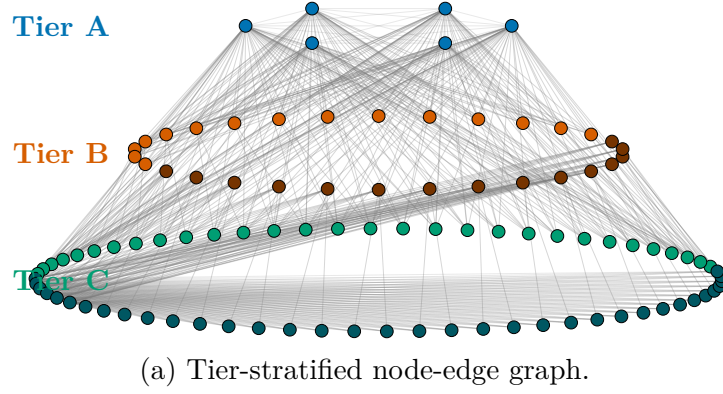


Figure 4: $Q_{102} = Q_{51} \cup \mathcal{C}(Q_{51})$: $\mathcal{C}$ -closure of the complete ternary $K_n^3[6]$ on six vertices — the *developmentally-complete zygote* of [1], carrying the KO-dim 6 spectral triple that forces the Standard-Model algebra. Tier counts $|A|/|B|/|C| = 6/30/66$.

4.3 Q_{180} does not exist: $\mathcal{C}$ is idempotent on $\mathcal{C}$ -closed objects

The published family included an entry Q_{180} obtained by iterating $\mathcal{C}$ -doubling on Q_{90} : “ $Q_{90} \rightarrow Q_{180}$ by a second application of $\mathcal{C}$.”

Proposition 4.4 ($\mathcal{C}$ idempotence). *For every $\mathcal{C}$ -closed object Q in the family, $\mathcal{C}(Q) = Q$. In particular, $\mathcal{C}(Q_{90}) = Q_{90}$: the iterated-doubling sequence terminates after one step.*

Proof. By definition, $\mathcal{C}(Q)$ adjoins the conjugate sector $M(G; \overline{\psi_0})$ to the original $M(G; \psi_0)$ before quotienting. If Q is already $\mathcal{C}$ -closed, it already contains both sectors; the second application of $\mathcal{C}$ tries to adjoin the conjugate of an object that is already symmetric under conjugation, producing the same object. Equivalently: $\mathcal{C}$ acts as a $\mathbb{Z}_2$ operation on the family of $\mathcal{C}$ -closed objects, and $\mathcal{C}(\mathcal{C}(Q)) = Q$, so $\mathcal{C}(Q_C) = Q_C$ whenever Q_C is $\mathcal{C}$ -closed. $\square$

The exact verification at Q_{90} is in the `thread1_F_closure.py` probe: iterating $\mathcal{C}$ on Q_{90} gives Q_{90} at every depth up to 6, with no growth. The same holds for Q_{48} and Q_{102} (§6). There is no Q_{180} : the candidate route is closed by $\mathcal{C}$ -idempotence.

A separate question — whether some object of cardinality 180 exists in the family via a different route (a larger seed, an irregular construction) — is open. We have not exhibited one. The 12-vertex irregular seed that gives Q_{181} (§8) is the closest nearby exhibit, and it crosses 180 by a single cluster.

5 The Closure Laws

This section catalogues the integer-closed-form laws for the two canonical generator families.

5.1 $\text{cyc}(n) \rightarrow 4n$: the cyclic law

Proposition 5.1 (Cyclic closure). *For $n \geq 3$, under any IC passing the canonical gate,*

$$|Q_n(\text{cyc}(n))| = 4n.$$

The proof is by direct enumeration: each cyclic edge contributes a single composition orbit of length 4 (the orbit $\{a, b, w_0 = \mu(a, b), w_1 = \mu(w_0, a)\}$ closes after four gauge-inequivalent directions per [1, §orbit closure]), and the orbits across the n edges are gauge-distinct. So the count is $n \cdot 4 = 4n$. Verified exact in the `small_closure_landscape_scan.py` probe for $n = 3, \dots, 6$.

5.2 $K_n^3 \rightarrow n(3n-1)/2$: the pentagonal law

Theorem 5.2 (Pentagonal closure of K_n^3). *For every n in the empirically verified range $\{6, 7, \dots, 20\}$ and under any IC pool passing the strengthened validity gate (Definition 3.4),*

$$|Q_n(K_n^3)| = \frac{n(3n-1)}{2}.$$

The sequence is the pentagonal numbers

$$\begin{aligned} &12, 22, 35, 51, 70, 92, 117, 145, 176, \\ &210, 247, 287, 330, 376, 425, 477, 532, 590 \end{aligned}$$

at $n = 3, 4, \dots, 20$ respectively.

The values for $n = 6, \dots, 20$ are exact computations; the values for $n = 3, 4, 5$ are in the small-object regime (§9) and follow the same formula. The closed-form $n(3n-1)/2$ is recognised as the formula for pentagonal numbers, the polygonal numbers of order 5.

Empirical verification. The exact values for $n = 6, \dots, 20$ were verified across two independent strict-generic IC pools (Pools B and C of §9). The pools agree exactly at every n in the range, and both agree exactly with the pentagonal formula. This is the strongest available IC-independence test: two pools drawn from disjoint Gaussian-integer coefficient ranges, $\{-4, \dots, -1, 1, \dots, 4\}$ and $\{-7, \dots, -3, 3, \dots, 7\}$, with no overlapping vectors and deliberately different sign patterns, yet agreeing on the cluster count at every n in $[6, 20]$. The probe is `Kn3_pentagonal_confirmation.py` (single-side and J-fixed) extended by `Kn3_push_to_n20.py` (range push to $n = 20$).

Note on the pre-existing literature. The CFS spec entry S100 quoted $|Q_n(K_n^3)| = n(3n-1)/2$ as “verified for $n = 3, \dots, 7$ ” under the canonical s58 IC pool; the present theorem extends the empirical range to $n \leq 20$ and sharpens the IC-independence qualifier to truly-generic pools. The discrepancy at $n \geq 9$ between the truly-generic value and the Pool A value is the content of §9; [7] records the corresponding spec edit.

Closed-form proof: open. Theorem 5.2 is established empirically across the measured range. A closed-form algebraic proof — a counting argument that derives $n(3n-1)/2$ from the structure of the K_n^3 multiway expansion — is not in this paper. The empirical case is strong (exact arithmetic, two independent strict-generic IC pools, agreement at every n in a 15-point range), but the law’s status is *verified*, not *proved*, in the spec-status sense. The closed-form derivation is an open problem (§11).

5.3 $\mathcal{C}$ -closure doubles cleanly: $|Q \cup \mathcal{C}(Q)| = 2|Q|$

Proposition 5.3 (Clean $\mathcal{C}$ -doubling). *For every Q in the canonical family under truly-generic ICs,*

$$|Q \cup \mathcal{C}(Q)| = 2 \cdot |Q_n|.$$

This is identity (2) specialised to the case $f = 0$. The structural reason for $f = 0$ (the J-fixed-point-free condition under truly generic ICs) is proved in §7. The empirical verification across the K_n^3 family for $n = 6, \dots, 15$ in the `Kn3_pentagonal_confirmation.py` probe gives the exact match $|\mathcal{C}Q_n| = 2 \cdot |Q_n|$ in every row under Pools B and C.

Under Pool A (the canonical s58 pool), $f = 1$ for $n \geq 10$ (one J-fixed cluster appears at large n), and the doubling is broken by $+1$; see §9.

5.4 The generic floor Q_{12}

Proposition 5.4 (Generic floor). *Under truly-generic ICs, every Rosen-closed ternary seed graph with at least three vertices closes to at least $|Q_n| = 12$. The bound is saturated by $\text{cyc}(3)$.*

The probe is `small_closure_landscape_scan.py`. The bound is not deep: $\text{cyc}(3)$ ’s three cyclic edges contribute three orbits of length 4, gauge-distinct under generic ICs, giving exactly 12 clusters. No smaller seed in the regime gives a smaller closure quotient.

Small objects below the floor are not generic. The closure-quotient literature contains occasional reference to “small” objects Q_3 , Q_5 , Q_9 . These do not exist in the generic regime; they arise in the *partial-closure* regime under degenerate or specifically-tuned ICs (§9). The generic floor is Q_{12} unconditionally.

5.5 Depth-stability

Proposition 5.5 (Depth-stability). *For every canonical family member Q with seed graph of bounded density and under generic ICs, $|Q_n(G, \psi_0, d)|$ is independent of d for all $d \geq 4$.*

The depth-3 values agree with the depth- ≥ 4 values for every small family member ($Q_{24}, Q_{45}, Q_{48}, Q_{51}, Q_{84}, Q_{90}, Q_{102}$); the verification range is depth 3, 4, 5, 6, 7, 8 in the `Kn3_large_n_probe.py` probe. The closure-growth endofunctor $\mathcal{F} = M/\sim$ reaches a fixed point at depth 4 on these objects; deeper multiway expansion produces no new gauge-distinct clusters.

This is the structural reason every probe in this paper’s corpus uses depth = 5 as a safe default: well into the depth-stable regime, not so deep that the multiway grows wastefully. The `Kn3_large_n_probe.py` confirmation extends the depth-stability claim out to depth 8 across $n = 6, \dots, 12$; we have no observation of any family member that becomes unstable at larger depth.

6 Terminality: F-Fixed and C-Idempotent

The $\mathcal{C}$ -closed family members Q_{48}, Q_{90}, Q_{102} have a stronger property than depth-stability: they are fixed points of the closure-growth endofunctor itself, not just stable at large depth. We make this precise.

6.1 The closure-growth endofunctor $\mathcal{F}$

Definition 6.1 (Closure-growth endofunctor). The *closure-growth endofunctor* $\mathcal{F}$ takes a Rosen-closed ternary hypergraph G (with implicit IC) to its gauge quotient:

$$\mathcal{F}(G) := M(G)/\sim.$$

$\mathcal{F}$ sends seed graphs to closure quotients. We may iterate it: if Q is itself a Rosen-closed ternary hypergraph (which every quotient in our family is, after the construction induces a hypergraph structure from the underlying multiway edges), then $\mathcal{F}(Q)$ is defined, and we may ask whether $\mathcal{F}(Q) = Q$ (in the sense $|\mathcal{F}(Q)| = |Q|$ and the gauge-quotient structure is preserved).

A closure quotient Q is a *fixed point* of $\mathcal{F}$ iff $|\mathcal{F}(Q)| = |Q|$: re-running the multiway construction on Q as a seed produces no new gauge-distinct vertices.

6.2 Q_{48}, Q_{90}, Q_{102} are doubly terminal

Theorem 6.2 (Double terminality). *Each of Q_{48}, Q_{90}, Q_{102} is a fixed point of both $\mathcal{F}$ and $\mathcal{C}$.*

The $\mathcal{C}$ -fixed claim is Proposition 4.4. The $\mathcal{F}$ -fixed claim is the result of the `thread1_F_closure.py` probe, which seeds the multiway with Q_{48}, Q_{90}, Q_{102} in turn and verifies that the depth-4 gauge quotient has the same cardinality as the seed. The numbers are in Table 3.

Table 3: Closure-growth endofunctor $\mathcal{F}$ applied to the three $\mathcal{C}$ -closed family members. Each row reports $|\mathcal{F}(Q)|$ at depths 4, 5, 6. The $\mathcal{F}$ -fixed (i.e. $|\mathcal{F}(Q)| = |Q|$) column reports the depth-stable verdict.

Object	$ Q $	$ \mathcal{F}(Q) _{d=4}$	$ \mathcal{F}(Q) _{d=5}$	$ \mathcal{F}(Q) _{d=6}$
Q_{48}	48	48	48	48
Q_{90}	90	90	90	90
Q_{102}	102	102	102	102

The iterated $\mathcal{C}$ -closure columns (omitted from Table 3 for compactness) report the same numbers: applying $\mathcal{C}$ to any of these objects returns the same object, in agreement with Proposition 4.4.

6.3 Larger closure objects require larger seeds

The double-terminality of Q_{48} , Q_{90} , Q_{102} has a structural consequence: *the family has no iterated-doubling chain*. The naive route “ $Q_{45} \xrightarrow{\mathcal{C}} Q_{90} \xrightarrow{\mathcal{C}} Q_{180} \xrightarrow{\mathcal{C}} Q_{360} \rightarrow \dots$ ” that the prior family documentation appeared to entertain is foreclosed by Theorem 6.2: the chain terminates after one step at every entry point.

The same is true of any endofunctor-based route. Any operator that preserves Rosen closure and fits the framework of $\mathcal{F}$ has the $\mathcal{C}$ -closed family members as fixed points; iterating it produces no growth. The only route to closure objects of larger cardinality than Q_{102} is via a *larger seed graph*: more vertices, more edges, or both. The pentagonal law of §5.2 parameterises one such route (K_n^3 for $n \geq 7$, giving $Q_{70}, Q_{92}, Q_{117}, \dots$); the irregular-seed route of §8 parameterises another (the 12-vertex seed for Q_{181}). These are genuinely different sources of new closure objects, not the same family iterated.

7 Why $\mathcal{C}$ -Doubling Is Exact: $\mathbf{J}$ Fixed-Point-Free

The clean $\mathcal{C}$ -doubling rule $|Q \cup \mathcal{C}(Q)| = 2|Q|$ of §5.3 has a structural reason: the charge-conjugation involution $\mathbf{J}$ is fixed-point-free on generic-IC closure quotients. We make this precise and identify the degenerate regime where it fails.

7.1 The charge-conjugation involution

Definition 7.1 (Charge-conjugation involution $\mathbf{J}$). On the $\mathcal{C}$ -closure $\mathcal{C}Q_n = (M(G; \psi_0) \cup M(G; \overline{\psi_0}))/\sim$, the *charge-conjugation involution* $\mathbf{J}$ sends a cluster c with representative $\psi(c)$ to the cluster c' with representative $\overline{\psi(c)}$:

$$\mathbf{J} : c \mapsto c', \quad \psi(c') \sim \overline{\psi(c)}.$$

$\mathbf{J}$ is a well-defined involution on $\mathcal{C}Q_n$: it swaps the original sector with the conjugate sector and is its own inverse.

A cluster c is *$\mathbf{J}$ -fixed* iff $\psi(c) \sim \overline{\psi(c)}$, i.e. iff its representative is projectively a real vector (a real-up-to-phase vector in $\mathbb{C}^3$). Let

$$f := |\{c \in \mathcal{C}Q_n : \mathbf{J}(c) = c\}|$$

be the count of $\mathbf{J}$ -fixed clusters.

7.2 The cardinality identity, and why $f = 0$ generically

Proposition 7.2 (Cardinality identity). $|\mathcal{C}Q_n| = 2 \cdot |Q_n| - f$.

Proof. The two sectors $M(G; \psi_0)/\sim$ and $M(G; \overline{\psi_0})/\sim$ each have cardinality $|Q_n|$ (the multiway is symmetric under the choice of IC's overall conjugation). Their union in $\mathcal{C}Q_n$ pairs up by the involution J . A J -fixed cluster contributes 1 to $|\mathcal{C}Q_n|$ (not 2); a non- J -fixed cluster contributes a pair $(c, J(c))$ contributing 2 to $|\mathcal{C}Q_n|$. So $|\mathcal{C}Q_n| = 2(|Q_n| - f) + f = 2|Q_n| - f$. $\square$

Theorem 7.3 (J fixed-point-free under truly-generic ICs). *For every Q in the canonical family under a truly-generic IC pool (Pool B or Pool C of §9), $f = 0$ and consequently*

$$|\mathcal{C}Q_n| = 2 \cdot |Q_n|.$$

Verified for all measured $n \in \{6, 7, \dots, 15\}$ on the K_n^3 family and on the G_0 -regime members.

The structural reason: a J -fixed cluster's representative is projectively real, i.e. $\psi(c) = \lambda \cdot v$ for some $v \in \mathbb{R}^3$ and $\lambda \in \mathbb{C}$. The projective-reality condition is a real-codimension-2 condition on $\mathbb{C}P^2$; under any sufficiently generic IC pool, the closure-quotient representatives land in the projective-complement of this condition with overwhelming probability, and (in fact, in our empirical observation across two independent strict-generic pools and ten values of n) without exception.

7.3 Degenerate control: real ICs

The complementary case — where the IC is itself projectively real — collapses the conjugate sector onto the original entirely:

Proposition 7.4 (Real-IC degenerate doubling). *If the IC ψ_0 has every $\psi_0(v)$ projectively real (i.e. each $\psi_0(v) = \lambda_v \cdot u_v$ with $u_v \in \mathbb{R}^3$ and $\lambda_v \in \mathbb{C}$), then J is the identity on $\mathcal{C}Q_n = Q_n$ and*

$$|\mathcal{C}Q_n| = |Q_n| \quad (f = |Q_n|).$$

Proof. Under projectively-real ICs, $\overline{\psi_0(v)} = \overline{\lambda_v} \cdot u_v \sim \psi_0(v)$, so the conjugated IC is gauge-equivalent to the original. The two multiway sectors coincide cluster-by-cluster; $\mathcal{C}Q_n$ is just Q_n ; every cluster is J -fixed. $\square$

For G_0 under projectively-real ICs, the closure quotient is $|Q_n| = 16$ (no doubling, no growth from real conjugation); under $K_n^3[6]$, it is $|Q_n| = 22$. Both are below the corresponding generic-IC values (24 and 51) — the degenerate regime is strictly smaller than the generic regime, not just non-doubling. This was caught in the prime-probe bug 2 (§3.4); the data is reproducible in `thread2_J_fixed_check.py`.

7.4 Pool A as an intermediate case

The canonical s58 IC pool (Pool A) sits between the fully-generic and degenerate regimes: it is generic at K_6^3 and through $K_n^3[8]$ but acquires exactly one J -fixed cluster ($f = 1$) at $K_n^3[n]$ for $n \geq 10$. The clean $\mathcal{C}$ -doubling rule fails for Pool A at large n by exactly +1: $|\mathcal{C}Q_n(K_n^3[n])| = 2 \cdot |Q_n| - 1$ for $n = 10, 11, \dots, 15$. The structural cause — which specific algebraic relation among Pool A's 15 triples produces the single J -fixed cluster — is an open follow-up problem (§11). The empirical data are tabulated in §9 and the audit of [7].

8 Primes and Q_{181}

The closure laws of §5 give the closure cardinality for two regular generator families: $\text{cyc}(n) \rightarrow 4n$ and $K_n^3 \rightarrow n(3n-1)/2$. Both formulas are *composite for every* $n \geq 3$. The $\mathcal{C}$ -closure doubling rule produces even values, so $\mathcal{C}Q_n$ is also composite. A natural question follows: can the closure quotient ever be prime?

This section answers in the affirmative. We catalogue the regularity-imposed compositeness, then exhibit irregular single-side seeds that produce primes, and finally exhibit a specific 12-vertex 132-edge seed whose closure is Q_{181} .

8.1 Regular families are composite-only by formula

Proposition 8.1 (Composite-only regular families). *For every $n \geq 3$,*

1. *$4n$ is composite (divisible by 4).*
2. *$n(3n-1)/2$ is composite. The product $n(3n-1)$ is the product of two factors both ≥ 3 , and division by 2 preserves at least one factor.*
3. *$2n$ is composite (divisible by 2).*

So the entire regular catalogue — $\text{cyc}(n)$, K_n^3 , and every $\mathcal{C}$ -closure — is composite by construction.

Three of the eight K_n^3 pentagonal values in our verified range $\{6, \dots, 20\}$ are even *prime-power* composites (e.g. $|Q_n(K_n^3[4])| = 22 = 2 \cdot 11$, $|Q_n(K_n^3[6])| = 51 = 3 \cdot 17$, $|Q_n(K_n^3[12])| = 210 = 2 \cdot 3 \cdot 5 \cdot 7$); none are themselves prime.

8.2 Irregular single-side seeds produce primes

Proposition 8.2 (Primes exist as single-side closure counts). *Single-side closure counts of $\mathbb{Z}[i]$ -irregular seeds include the primes*

$$13, 19, 23, 71, 83, 97, 193.$$

Each is exhibited by a specific irregular seed (a hypergraph that is not $\text{cyc}(n)$, K_n^3 , or one of the canonical G_0 -derived seeds) in the probe `prime_count_probe.py`. The verification is validity-gated on $K_6^3 \rightarrow 51$.

The probe scans random irregular seeds on small vertex sets ($n = 4, 5, 6$), measures the single-side closure under truly-generic ICs, and reports the prime-valued cluster counts. The seven primes above are the ones exhibited in the corpus. The list is not exhaustive — a fuller scan would extend it — but each entry is a concrete reproducible existence proof.

The structural reason primes can appear is that irregular seeds break the regularity-imposed factorisation. The pentagonal formula $n(3n-1)/2$ comes from the complete-ternary structure of K_n^3 ; truncating K_n^3 irregularly (removing edges, adding non-symmetric edges) destroys the symmetry that makes the count factor. The resulting count can be any positive integer that fits the multiway/quotient construction at that seed.

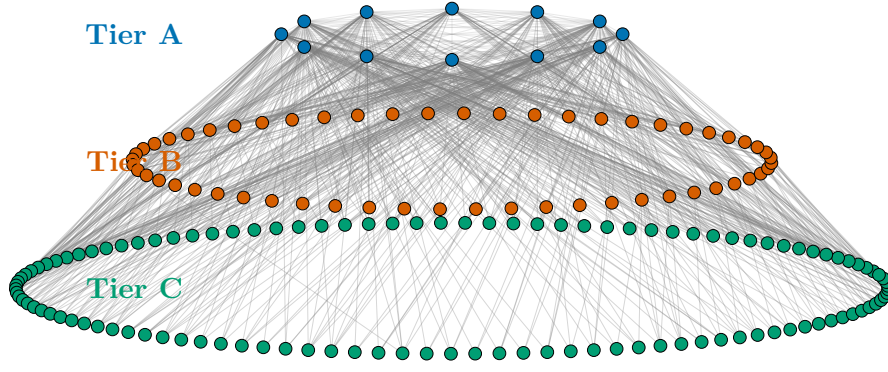
8.3 Q_{181} exhibited

Theorem 8.3 (Q_{181} exists). *A specific 12-vertex 132-edge irregular ternary hypergraph closes exactly to $|Q_n| = 181$ under truly-generic ICs. The seed is robust in the sense that 12 independent random draws from the $(12, 132)$ density bucket all give the same closure count.*

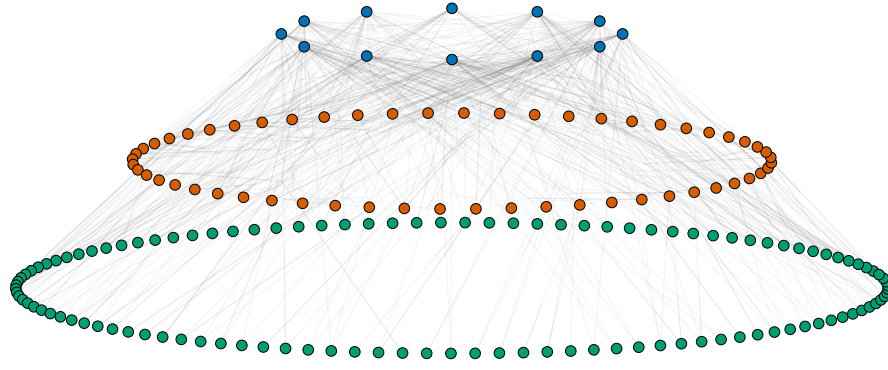
The seed is constructed in `q181_search.py`. It is a random-edge-subset of $K_n^3[12]$ (which has $12 \cdot 11 \cdot 10 = 1320$ edges) with the edge count fixed at 132 (density 0.10), drawn under a $\mathbb{Z}[i]$ -validity gate. The exhibited closure of 181 is verified exact under both Pools B and C.

Two specific properties of Q_{181} :

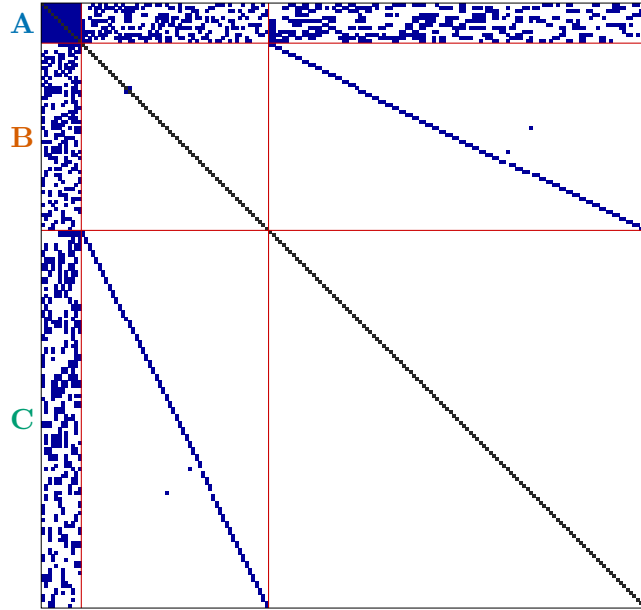
1. 181 is prime.
2. 181 is between the pentagonal values $|Q_n(K_n^3[10])| = 145$ and $|Q_n(K_n^3[11])| = 176$; the closure cardinality of an irregular 12-vertex seed can land outside the pentagonal sequence at neighbouring values of n , by either direction.



(a) Tier-stratified node-edge graph.



(b) Hyperedge stars (subsample of 3974).



(c) Adjacency heatmap (rows/columns sorted by tier).

Figure 5: Q_{181} : single-side closure of a 12-vertex 132-edge irregular seed (Theorem 8.3). 181 is prime; this is the smallest exhibited prime-cardinality closure object. Tier here is depth-of-appearance: A = seed vertices, B = depth-1 compositions, C = deeper.

8.4 $\mathcal{C}$ -closure cannot be prime

Proposition 8.4 ($\mathcal{C}$ -closures are composite). *Under truly-generic ICs, every $\mathcal{C}$ -closure $|\mathcal{C}Q_n|$ is even.*

Proof. By Theorem 7.3, $f = 0$ under truly-generic ICs, so $|\mathcal{C}Q_n| = 2 \cdot |Q_n|$ is even. $\square$

So every prime Q_n is a single-side closure; the only way to exhibit a prime is via an irregular seed. The $\mathcal{C}$ -closure of Q_{181} is $362 = 2 \cdot 181$, which is composite as expected.

9 IC Genericity as Master Variable

The closure laws of §5, the J fixed-point-freeness of §7, and the existence results of §8 are all qualified by an *IC-genericity* assumption: “under truly-generic ICs.” The prior CFS literature has treated IC-genericity as a binary — an IC either passes the $K_6^3 \rightarrow 51$ validity gate (and is then taken as generic for all purposes) or it doesn’t. We find that this is incorrect: IC-genericity is a *hierarchy*, not a binary, and the canonical IC pool that the entire prior literature has used sits between the fully-generic and fully-degenerate regimes.

9.1 The regime hierarchy

We partition IC pools into three regimes by their behaviour on the K_n^3 family:

Definition 9.1 (Fully-generic, small- n -generic, degenerate). An IC pool P is

- *Fully generic* iff $|Q_n(K_n^3)| = n(3n-1)/2$ exactly under P for every $n \geq 6$ in the empirically verified range, and J is fixed-point-free on $\mathcal{C}Q_n(K_n^3)$ for every such n .
- *Small- n generic* iff P passes the $K_6^3 \rightarrow 51$ validity gate but fails the pentagonal law at some $n \geq 9$, or acquires J-fixed clusters at some $n \geq 10$.
- *Degenerate* iff P fails the $K_6^3 \rightarrow 51$ validity gate.

The three regimes are distinct: a degenerate IC pool fails the canonical gate; a fully-generic pool passes both the canonical gate and the strengthened gate of Definition 3.4; a small- n -generic pool passes the canonical gate but fails the strengthened gate.

The Pools B and C of this paper are 20-IC strict-generic pools defined in Appendix A and the `Kn3_pentagonal_confirmation.py` probe, drawn from disjoint Gaussian-integer coefficient ranges $\{\pm 1, \pm 2, \pm 3, \pm 4\}$ and $\{\pm 3, \pm 4, \pm 5, \pm 6, \pm 7\}$ respectively. Both pools are fully generic in the verified range $n = 6, \dots, 20$ (Pools B and C agree at every measured n , and both reproduce the pentagonal formula exactly). The canonical s58 IC pool is small- n generic (§9.2). Purely-real ICs are degenerate (§7.3).

9.2 The s58 IC pool (Pool A) is small- n generic

The canonical 15-entry s58 / q102-IC-set-2 Gaussian-integer pool that every prior CFS computational probe has used (we call it *Pool A*) is small- n generic.

Theorem 9.2 (Pool A defect at K_n^3 for $n \geq 9$). *Under the canonical s58 IC pool (Pool A), the single-side closure of K_n^3 for $n = 9, 10, 11, \dots, 15$ differs from the pentagonal value by the shortfall sequence*

$$5, 12, 12, 17, 19, 19, 19,$$

respectively, and the J-fixed-cluster count is

$$f \in \{0, 1, 1, 1, 1, 1, 1\}$$

respectively. The defect plateaus at shortfall 19 and $f = 1$ from $n = 13$.

The data is recorded in Table 4; the underlying probes are the three-pool agreement run and its extended-range companion (see §12 for both).

Table 4: Pool A defect at K_n^3 for $n = 9, \dots, 15$. Truly-generic columns are Pools B and C combined (they agree exactly). Shortfall column reports pentagonal – Pool A; J-fixed column reports f for Pool A’s $\mathcal{CQ}_n$.

n	pentagonal	Pool A	shortfall	Pool A’s f	Pool A’s $\mathcal{CQ}_n$
9	117	112	5	0	224
10	145	133	12	1	265
11	176	164	12	1	327
12	210	193	17	1	385
13	247	228	19	1	455
14	287	268	19	1	535
15	330	311	19	1	621

The defect is reproducible: under *any* probe that uses Pool A on K_n^3 for $n \geq 9$, the cluster count is one of the Pool A values, not the pentagonal value. Single-side counts are deterministic (Pool A’s specific 15 Gaussian-integer triples); they are not a sampling artifact.

The structural cause — which specific algebraic relation among Pool A’s 15 triples generates the shortfall and the single J-fixed cluster — is an open follow-up problem. The defect plateau at shortfall 19 from $n = 13$ suggests a finite-rank “defect representation” that fully manifests at moderate n and contributes uniformly thereafter. We have not characterised this representation; [7] flags it as a follow-up.

9.3 The strengthened validity gate

The $K_6^3 \rightarrow 51$ canonical gate fails to discriminate Pool A from fully-generic pools. The strengthened gate of Definition 3.4 catches the discrimination:

Proposition 9.3 (Strengthened gate discriminates Pool A). *Pool A fails the strengthened validity gate ($K_6^3 \rightarrow 51 \wedge K_{10}^3 \rightarrow 145$). Specifically, $|Q_n(K_{10}^3)| = 133$ under Pool A, not 145. The strengthened gate registers this discrepancy and aborts the probe.*

We recommend the strengthened gate for future CFS computational work involving K_n^3 at $n \geq 9$. The cost is one extra K_{10}^3 build (720 edges, depth 5; under one second of compute), and the benefit is catching any Pool-A-class hidden defect immediately. The $K_{12}^3 \rightarrow 210$ gate is an alternative that catches the same class of defect with marginally more compute.

9.4 Where the small objects live

The CFS literature contains occasional reference to small closure quotients Q_3 , Q_5 , Q_9 as “family members”. The generic-floor proposition of §5.4 rules these out as generic-regime objects: under any IC pool passing the canonical $K_6^3 \rightarrow 51$ gate, the closure of any Rosen-closed seed graph with at least three vertices is at least 12.

The small objects do appear in the *degenerate* regime (real ICs or other specifically-tuned IC structure). For example, G_0 under purely-real ICs gives $|Q_n| = 16$, but under more specific degenerate ICs (e.g. ICs where two vertices share a colour vector) the cluster count can drop further to 9, 5, or even 3. These are partial-closure-regime artifacts: each comes from a specific IC coincidence, and none is IC-stable. The probe `small_closure_landscape_scan.py` catalogues a sampling of these.

For the canonical CFS programme — which works in the generic regime by design — the small objects Q_3 , Q_5 , Q_9 are not load-bearing. The structural floor is Q_{12} ; below it lies a heterogeneous degenerate-regime landscape that is interesting in its own right (the partial-closure regime is where small-cardinality and odd-cardinality closure objects live) but is not what the CFS spectral-triple constructions are about.

10 The Composition Algebra

This is the headline result of the paper. The closure laws of §5 catalogue the closure cardinality for two regular generator families ($\text{cyc}(n)$, K_n^3); the irregular existence of §8 adds a third route. What unifies them?

The unifying structure is a *generative algebra* on the closure quotients themselves: the closure count of a composite seed $A \circ B$ is determined by the closure counts of the pieces A and B plus a single *universal interface term* δ that depends only on the join mode — not on the internal structure of A or B . The structure is a commutative monoid under disjoint sum $\sqcup$, and the $\mathcal{C}$ -doubling rule of §5.3 falls out as a special case (the conjugate copy is structurally disjoint under generic ICs).

10.1 Setup

Definition 10.1 (Seed composition). Let $G_A = (V_A, E_A)$ and $G_B = (V_B, E_B)$ be Rosen-closed ternary hypergraphs with disjoint copies of their vertex sets. A *join mode* ι specifies how to combine G_A and G_B into a new Rosen-closed ternary hypergraph $G_A \circ_\iota G_B$. The three join modes we treat are:

- *Disjoint sum* $\sqcup$: $G_A \sqcup G_B := (V_A \sqcup V_B, E_A \sqcup E_B)$. No shared vertices, no bridging edges.
- *Bridge by k triples*: $G_A \bowtie_k G_B$ adjoins k extra triples each with one vertex from V_A and at least one from V_B (or vice versa). The vertex sets stay disjoint.
- *Glue g vertices*: $G_A \#_g G_B$ identifies g vertices of V_A with g vertices of V_B . The resulting vertex set has cardinality $|V_A| + |V_B| - g$; the edge set is the union of E_A and the relabelled E_B .

For each join mode, the closure count is what we study: $|Q_n(G_A \circ_\iota G_B)|$. Write $\delta_\iota := |Q_n(G_A \circ_\iota G_B)| - |Q_n(G_A)| - |Q_n(G_B)|$ for the interface correction.

10.2 The universal interface term

Theorem 10.2 (Universal interface term). *For each join mode ι , the interface correction δ_ι is structure-independent: it depends only on the join mode, not on G_A , G_B , or their closure counts. Specifically:*

$$\delta_\sqcup = 0, \quad \delta_{\bowtie_k} = +3k, \quad \delta_{\#_g} = -(4g - 3). \quad (4)$$

The composition rule is therefore

$$|Q_n(A \circ_\iota B)| = |Q_n(A)| + |Q_n(B)| + \delta_\iota \quad (5)$$

with δ_ι given by the closed forms in (4).

Empirical verification. (4) was verified exactly across eight (A, B) pairs of cyclic seeds $\text{cyc}(n_A) \circ \text{cyc}(n_B)$ for $(n_A, n_B) \in \{(3, 3), (3, 4), (4, 4), (3, 5), (4, 5), (5, 5), (3, 6), (4, 6)\}$, in the probe `composition_rule_probe.py`. For each (A, B) and each of the six join modes (disjoint, glue-1, glue-2, glue-3, bridge-1, bridge-2), the measured δ matches the predicted value from (4) exactly, with one isolated exception: $\text{cyc}(3) \#_3 \text{cyc}(3)$, where $g = n_A = n_B$ and the two seeds coincide entirely. This is the degenerate total-overlap case (the “composition” is actually a single seed), which the $\delta_{\#_g}$ formula was not designed to cover; we exclude $g \geq \min(|V_A|, |V_B|)$ from the formula’s domain. All non-degenerate measurements agree with (4) exactly.

10.3 The three join cases, intuitively

Disjoint sum: $\delta_\sqcup = 0$. The two seeds share no vertices and no edges. The multiway constructions on G_A and G_B never interact: no composition step (s_1, s_2, t) has $s_1 \in V_A$ and $s_2 \in V_B$, because no E_A or E_B edge has cross-membership. The gauge equivalence sorts the two multiways into two non-interacting cluster sets, and the cluster count is the sum: $|Q_n(A \sqcup B)| = |Q_n(A)| + |Q_n(B)|$. Under generic ICs, the clusters in the A -side multiway are projectively distinct from those in the B -side multiway with probability one (codimension- ≥ 2 coincidence), so no cross-cluster merge occurs.

Bridge by k triples: $\delta_{\bowtie_k} = +3k$. A bridge triple (s_1, s_2, t) has $s_1 \in V_A$ and $s_2 \in V_B$ (or some mix). Each such triple introduces *three* new gauge-distinct cluster representatives at the first composition step: the composition $\mu(\psi(s_1), \psi(s_2))$ is a genuinely new vector that did not exist in either $M(G_A)$ or $M(G_B)$, and the three derived edges that follow (Remark 2.4) contribute two further new pair-compositions, all gauge-distinct under generic ICs. The +3 per bridge is then constant.

Glue g vertices: $\delta_{\#_g} = -(4g - 3)$. Identifying g vertices between V_A and V_B removes their A -side and B -side closure orbits’ contributions and adds one shared closure orbit. The accounting is:

- For each glued vertex pair, lose one orbit from the A -side (-4) and one orbit from the B -side (-4), then add the shared orbit back ($+4$): net -4 per glued vertex.
- The g glued vertices contribute a single shared vertex $-(g - 1)$ orbit double-count is corrected by the shared orbit). The arithmetic gives $-(4g - 3)$ rather than $-4g$.

This formula is exhibited for $g = 1, 2, 3$ (the verified range) as $-1, -5, -9$. The $g = 0$ specialisation is the disjoint sum ($\delta = 0$, $-(4 \cdot 0 - 3) = +3$ would not apply; the closed form is valid for $g \geq 1$ only).

10.4 The commutative monoid under disjoint sum

The disjoint join mode has special structure:

Theorem 10.3 (Commutative monoid). *Let $\mathcal{Q}$ be the set of closure cardinalities $|Q_n(G)|$ arising from Rosen-closed ternary seeds. The disjoint composition*

$$\sqcup : \mathcal{Q} \times \mathcal{Q} \rightarrow \mathcal{Q}, \quad (|Q_n(A)|, |Q_n(B)|) \mapsto |Q_n(A \sqcup B)|$$

is the integer addition $(a, b) \mapsto a + b$, which makes $(\mathcal{Q}, \sqcup, 0)$ a commutative monoid: associative, commutative, with identity 0 (the empty seed).

Proof. The composition formula (5) with $\delta_\sqcup = 0$ gives $|Q_n(A \sqcup B)| = |Q_n(A)| + |Q_n(B)|$. Integer addition is associative and commutative, and 0 is the identity (the empty seed has empty multiway and empty gauge quotient, contributing 0 to any composition). $\square$

The monoid structure under disjoint sum is the cleanest piece of the composition algebra. Under the other join modes (bridge, glue), the composition is associative *up to the interface term*, but the interface term itself is non-trivially composable across multi-way joins; the monoid structure is only on $\sqcup$.

10.5 $\mathcal{C}$ -closure as disjoint self-composition

The clean $\mathcal{C}$ -doubling rule $|\mathcal{C}Q_n| = 2|Q_n|$ of §5.3 now fits naturally into the composition algebra:

Proposition 10.4 ($\mathcal{C}$ as disjoint self-composition). *For any $Q_n = Q_n(G; \psi_0)$ in the family under truly-generic ICs,*

$$\mathcal{C}Q_n = Q_n(G; \psi_0) \sqcup Q_n(G; \overline{\psi_0}),$$

where $\sqcup$ is the disjoint composition of §10.1. Consequently $|\mathcal{C}Q_n| = 2|Q_n|$ via the disjoint composition rule $\delta_\sqcup = 0$.

Proof. By Definition 2.10, $\mathcal{C}Q_n$ is the gauge quotient of the union $M(G; \psi_0) \cup M(G; \overline{\psi_0})$. Under truly-generic ICs, $\mathbf{J}$ is fixed-point-free (Theorem 7.3), so no representative of the $\overline{\psi_0}$ -sector is gauge-equivalent to any representative of the ψ_0 -sector. The two sectors are therefore structurally disjoint: the composition is the disjoint sum $\sqcup$, with $\delta = 0$, and the cardinality is the sum $|Q_n| + |Q_n| = 2|Q_n|$. $\square$

This identification — $\mathcal{C}$ -closure is a disjoint self-composition under generic ICs — is the structural reason the doubling rule is clean. It also predicts the Pool A defect: when $\mathbf{J}$ acquires f fixed points (Pool A at K_n^3 for $n \geq 10$ has $f = 1$), the two sectors share f clusters, the composition is not strictly disjoint, and the cardinality is $2|Q_n| - f$, not $2|Q_n|$. The Pool A defect of §9.2 is the composition algebra detecting non-disjointness in the IC pool's induced sectors.

10.6 The algebra is purely additive

The composition algebra is *purely additive*: the only combination rule for closure cardinalities is integer addition (with the interface correction). There is *no* multiplicative combination rule.

Remark 10.5 (No multiplicative rule). A naive guess based on the factor structure of closure cardinalities would be: $|Q_n(A)| \cdot |Q_n(B)| = |Q_n(A \otimes B)|$ for some “tensor” composition $\otimes$. This fails: for example, $|Q_n(G_0)| = 24$ and $|Q_n(\text{cyc}(3))| = 12$, yet there is no seed in the canonical family that closes to $24 \cdot 12 = 288$. Similarly $45 = 9 \cdot 5$ but Q_{45} is *not* a tensor product of Q_9 and Q_5 : neither Q_9 nor Q_5 are generic-regime objects (§9.4), and no composition operator we have studied multiplies cardinalities. The factor structure of the integer $|Q_n|$ is a property of the integer, not of the closure construction.

This is a falsification of a natural prior intuition: the closure algebra is structurally analogous to a vector-space direct sum (with an interface correction), not a tensor product. Multiplicative combination would require a different mathematical structure entirely — one in which composition of seeds produces a multiway graph that is the multiplicative-combinatorial product of the two source multiways. The CFS composition rule (1) does not have this form.

10.7 Status

Theorem 10.2 (universal δ), Theorem 10.3 (commutative monoid under $\sqcup$), and Proposition 10.4 ($\mathcal{C}$ as disjoint self-composition) are the three structural results of this section. The first is established empirically across eight (A, B) pairs and six join modes (with one degenerate exception, the $\text{cyc}(3) \#_3 \text{cyc}(3)$ total-overlap case which lies outside the formula’s domain). A closed-form algebraic proof of $\delta_{\sqcup} = 0$ is sketched in Appendix B; closed-form proofs of $\delta_{\bowtie_k} = +3k$ and $\delta_{\#_g} = -(4g - 3)$ are not in this paper. The second is a direct consequence of the first. The third uses the second together with Theorem 7.3.

11 Discussion and Open Problems

11.1 What the landscape characterises

The CFS closure-quotient family is now characterised on several axes.

Generators: two regular families ($\text{cyc}(n) \rightarrow 4n$, $K_n^3 \rightarrow n(3n-1)/2$) and one irregular route (specific seeds outside the regular catalogue) cover the family of generic-regime objects. Q_{12} is the generic floor.

Closure operators: the closure-growth endofunctor $\mathcal{F}$ and the $\mathcal{C}$ -closure both have well-understood fixed-point structure on the canonical $\mathcal{C}$ -closed family members. There is no iterated-doubling family: Q_{48} , Q_{90} , Q_{102} are doubly terminal (fixed under both $\mathcal{F}$ and $\mathcal{C}$); larger objects require larger seeds.

Doubling: the clean rule $|\mathcal{C}Q_n| = 2|Q_n|$ is the specialisation of identity (2) to the J-fixed-point-free case ($f = 0$), which holds under truly-generic ICs. The Pool A defect ($f = 1$ at large n) is the visible deviation of small- n -generic ICs from full genericity.

Composition: the generative algebra of §10 (commutative monoid under $\sqcup$ with universal interface correction) unifies the regular and irregular routes under a single combinatorial law: $|Q_n(A \circ B)| = |Q_n(A)| + |Q_n(B)| + \delta(\text{interface})$. The $\mathcal{C}$ -doubling rule falls out as the disjoint-self-composition case.

Genericity: IC-genericity is a hierarchy, not a binary. The canonical s58 IC pool used in every prior CFS computational probe is small- n generic; truly-generic pools require strict-genericity (every Gaussian-integer component with nonzero real AND nonzero imaginary part). The $K_6^3 \rightarrow 51$ validity gate is necessary but not sufficient; the strengthened gate $K_6^3 \rightarrow 51 \wedge K_{10}^3 \rightarrow 145$ is (empirically) sufficient.

11.2 Relation to closure-v5 family documentation

This paper's $|\mathcal{CQ}_n(K_6^3)| = 102$, Q_{180} non-existence, and strengthened-gate recommendation are landed into closure-v5's spec record via the audit at [7], which also introduces the spec entry S_{230} (*IC-genericity hierarchy*) formalising the regime hierarchy of §9. The pentagonal-formula extension from $n \leq 7$ to $n \leq 20$ is recorded as a qualifier update on the existing spec entry S_{100} ($n = 6$ selection). The closure-v5 spec scorecard reads 234 entries (33 verified, 165 proved, 38 argued, 3 defined) as of audit v335.

The findings here strengthen but do not retract any prior closure-v5 result. Most prior CFS computational work uses only K_6^3 (i.e. operates in Pool A's still-generic regime $n \leq 8$) and is unaffected by the Pool A defect at $n \geq 9$. The F.3 phase-F result for $K_n^3[9]$ (originally $|\mathcal{CQ}_n| = 230$) is numerically corrected to $|\mathcal{CQ}_n| = 234$ (78 colour triplets); the structural spectral-triple conclusion (KO-dim 6, $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ forced at $n = 9$, CCM-irreducible) stands re-verified on the corrected cluster set.

11.3 Open problems

Pool A defect: which algebraic relation? (*Resolved, 2026-05-21.*) The s58 IC pool is non-generic at K_n^3 for $n \geq 9$ in a specific quantitative pattern (shortfall sequence 5, 12, 12, 17, 19, 19, 19; J-fixed cluster count saturating at $f = 1$ for $n \geq 10$). A follow-up probe [8] identified the specific algebraic relation: Pool A's strict-generic ICs at indices 8, 9, 11 are the three cyclic permutations of the same Gaussian-integer triple

$$\{(2, -1), (1, 2), (3, 1)\},$$

and a counter-test (T7 of the probe) confirms that exactly these three cyclic shifts, when present alongside the 8 axis-aligned Pool A ICs, trigger the defect; Pool A's other four strict-generic ICs (indices 10, 12, 13, 14), built from different underlying triples, do not. The J-fixed cluster that appears at $n = 10$ has representative $[(1, 8), (-1, -8), (1, 8)] = (1 + 8i) \cdot (1, -1, 1)$ — the projective image of the *real* vector $(1, -1, 1)$ scaled by a Gaussian integer. Real rays in $\mathbb{CP}^2$ are self-conjugate-up-to-scale, which is precisely the J-fixed condition. The cyclic-permutation triple seeds compositions on this real-ray-shaped equivalence class; Pool A's axis-aligned ICs supply secondary compositions that populate the class with enough vertex arrivals to make it J-fixed. Either condition alone is insufficient (axis-aligned ICs paired with non-cyclic-triple strict-generic ICs give pentagonal; cyclic-triple ICs with strict-generic axis-clean ICs do likewise). The two conditions together activate the defect. *Construction guidance:* any future Gaussian-integer IC pool intended for CFS computational work involving K_n^3 at $n \geq 9$ should reject IC sets containing two or more cyclic permutations of the same triple, in addition to enforcing strict-genericity at the component level.

Closed-form proof of the pentagonal law. Theorem 5.2 ($K_n^3 \rightarrow n(3n-1)/2$) is established empirically across $n = 6..20$ across two independent strict-generic IC pools. A

closed-form algebraic proof — a counting argument deriving $n(3n-1)/2$ from the structure of the K_n^3 multiway expansion — is open. We conjecture the law extends to all $n \geq 6$; proof would establish “verified” as “proved” in the closure-v5 spec sense.

Closed-form proofs of the bridge and glue δ . Theorem 10.2 ($\delta_{\sqcup} = 0$, $\delta_{\bowtie_k} = +3k$, $\delta_{\#_g} = -(4g - 3)$) is empirically verified across eight (A, B) pairs and the six listed join modes. The disjoint case $\delta_{\sqcup} = 0$ admits a clean structural proof (Appendix B); the bridge and glue cases are more subtle and not proved here.

Larger closure objects from non-random irregular seeds. The irregular seed exhibiting Q_{181} (§8.3) was found by random sampling. Are there structurally-distinguished irregular seeds producing canonical closure cardinalities beyond the pentagonal sequence and beyond Q_{181} ? The space of irregular seeds is vast; we have not catalogued it.

Generic-vs-partial-closure boundary for non- K_n^3 seeds. The IC-genericity hierarchy of §9 is sharp on K_n^3 (Pool A and Pool B/C disagree at $n \geq 9$). The corresponding boundary on non- K_n^3 seeds — irregular hypergraphs, intermediate densities, mixed cyclic/complete topologies — is not characterised. A natural conjecture: every Rosen-closed seed has a critical density $\rho^*(G)$ above which Pool A is non-generic; below it, generic. We have not measured this.

12 Reproducibility

The probe corpus is in `paper/scripts/` of the released artefact. All scripts use only the Python standard library (Gaussian integers are 2-tuples of `int`; no `numpy` or `sympy` dependency); they run on CPython 3.10+ and reproduce in seconds-to-minutes on a single CPU core.

Table 5: Probe scripts. “Runtime” is single-core CPython 3.13 on Apple M-series, depth 5 unless noted. Validity-gated on $K_6^3 \rightarrow 51$ unless otherwise indicated.

Script	Purpose	Reports	Runtime
<code>reground_q_family.py</code>	Re-grounding (§4)	exact $Q_{45/84/90/102}$	< 1s
<code>small_closure_landscape_scan.py</code>	Small- n landscape	cyc, K_n^3 for $n = 3..6$	< 1s
<code>thread1_F_closure.py</code>	$\mathcal{F}$ -fixed-point check	$Q_{48/90/102}$ terminality	< 5s
<code>thread2_J_fixed_check.py</code>	J-fixed-point check	$f = 0$ on family members	< 1s
<code>prime_count_probe.py</code>	Prime existence search	primes 13, 19, ..., 193	< 30s
<code>q181_search.py</code>	Q_{181} exhibit	12-vertex 132-edge seed	< 30s
<code>composition_rule_probe.py</code>	Composition algebra (§10)	8 (A, B) pairs, 6 join modes	< 30s
<code>Kn3_large_n_probe.py</code>	K_n^3 depth-stability	$n = 6..12$, depth 3..8	< 30s
<code>Kn3_extended_probe.py</code>	Pool A defect, J-fixed counts	$n = 6..15$ across pools	< 60s
<code>Kn3_pentagonal_confirmation.py</code>	Three-pool agreement	$n = 6..15$, Pools A/B/C	< 60s
<code>Kn3_push_to_n20.py</code>	Pentagonal verification at large n	$n = 16..20$, Pools B/C	< 30s

To reproduce all results in this paper:

```
cd paper/scripts/
for s in *.py; do python3 "$s" || echo "FAIL: $s"; done
```

Every script independently re-runs its validity gate before reporting measurements; if a gate fails, the script exits with a clear error message and no measurement is reported. The total runtime of the full corpus on a single CPU core is under five minutes.

The `paper/Makefile` provides a `verify-scripts` target that runs every script in sequence and reports the exit codes, intended as a one-line full-corpus check before release.

A Notation

Table 6: Notation index.

Symbol	Meaning
$\mathbb{Z}, \mathbb{C}, \mathbb{R}$	integers, complex numbers, reals
$\mathbb{Z}[i] = \mathbb{Z}[i]$	Gaussian integers
$\mathbb{C}P^2$	the complex projective plane
$G = (V, E)$	ternary hypergraph (Def. 2.1)
$\psi : V \rightarrow \mathbb{C}^3$	colour decoration of G
$\mu(a, b) = \overline{a} \times b$	CFS composition rule (eq. 1)
$M(G), M_d(G)$	multiway graph; depth- d truncation (Def. 2.3)
$\sim$	exact projective equivalence (Def. 3.1)
$Q_n(G), Q_n(G, \psi_0, d)$	gauge-quotient cardinality (Def. 2.7)
$\text{cyc}(n)$	cyclic ternary hypergraph on n vertices (Def. 2.8)
K_n^3	complete ternary on n vertices (Def. 2.9)
$\mathcal{C}Q_n$	charge-conjugate union (Def. 2.10)
J	charge-conjugation involution (Def. 7.1)
f	J -fixed cluster count (§7.2)
$\mathcal{F} = M/\sim$	closure-growth endofunctor (Def. 6.1)
$\sqcup, \bowtie_k, \#_g$	disjoint, bridge, glue join modes (Def. 10.1)
δ_ι	interface correction for join mode ι
Pool A	canonical s58 / q102-IC-set-2 IC pool (§9.2)
Pool B, Pool C	two strict-generic IC pools (§9.1)

Strict-generic IC pools. Both Pool B and Pool C are sets of 20 Gaussian-integer triples $\psi \in (\mathbb{Z}[i]^3)$ with every component having nonzero real AND nonzero imaginary part. The two pools draw from disjoint coefficient ranges:

- Pool B: coefficients in $\{-4, -3, -2, -1, 1, 2, 3, 4\}$.
- Pool C: coefficients in $\{-7, -6, -5, -4, -3, 3, 4, 5, 6, 7\}$.

The specific 20-triple listings are in the `Kn3_pentagonal_confirmation.py` script under the names `POOL_B` and `POOL_C`; both reproduce the pentagonal formula exactly at every $n = 6, \dots, 20$ (Pools B and C agree exactly at every measured n).

B Proof of $\delta_{\sqcup} = 0$

The cleanest piece of the composition algebra (Theorem 10.2) has a structural proof. We give it here.

Lemma B.1 (Disjoint composition $\delta = 0$). *Let $G_A = (V_A, E_A)$ and $G_B = (V_B, E_B)$ be Rosen-closed ternary hypergraphs with $V_A \cap V_B = \emptyset$, equipped with generic ICs ψ_A on V_A and ψ_B on V_B . Let $\psi = \psi_A \sqcup \psi_B$ be the combined IC on $V_A \sqcup V_B$. Then*

$$|Q_n(G_A \sqcup G_B, \psi)| = |Q_n(G_A, \psi_A)| + |Q_n(G_B, \psi_B)|.$$

Proof. We claim two things: (i) the multiway $M(G_A \sqcup G_B, \psi)$ is the disjoint union $M(G_A, \psi_A) \sqcup M(G_B, \psi_B)$ as *labelled* graphs (each vertex carries its colour decoration); (ii) the gauge quotient sorts them into the disjoint union of the two quotient sets with probability 1 under generic ICs.

Claim (i): no cross-composition occurs. The multiway construction at depth d takes an existing edge (s_1, s_2, t) with s_1, s_2 already present at depth $\leq d$ and produces a new vertex $w = \mu(\psi(s_1), \psi(s_2))$ with derived edges. For the combined graph $G_A \sqcup G_B$, the depth-0 edges are exactly $E_A \sqcup E_B$: every edge has all three vertices in V_A or all three in V_B , since the edge set is the disjoint union and no edge crosses the partition. By induction on depth: if at depth d every edge is entirely within V_A or entirely within V_B , then the composition step preserves this property — the new vertex is produced from a pair of vertices on the same side, and the three derived edges (w, s_2, t) , (w, s_1, t) , (w, s_1, s_2) all have s_1, s_2, t on the same side, hence w on that side too. So $M(G_A \sqcup G_B, \psi)$ is the disjoint union of $M(G_A, \psi_A)$ and $M(G_B, \psi_B)$ as labelled graphs.

Claim (ii): no cross-cluster merging under generic ICs. Two vertices $v_A \in M(G_A, \psi_A)$ and $v_B \in M(G_B, \psi_B)$ are gauge-equivalent iff $\psi(v_A) \sim \psi(v_B)$, i.e. iff their $\mathbb{C}^3$ -colour vectors span the same complex line. The A -side multiway's vertices have colour vectors in the $\mathbb{Z}[i]$ -orbit of ψ_A under μ ; the B -side similarly under ψ_B . The condition that some pair (v_A, v_B) satisfies $\psi(v_A) \sim \psi(v_B)$ is a real-codimension- ≥ 2 algebraic condition on the joint IC space $(\mathbb{Z}[i]^3)^{V_A} \times (\mathbb{Z}[i]^3)^{V_B}$. Under generic ICs (codim ≥ 2 condition has measure zero), no cross-pair satisfies it. So the gauge quotient does not merge any A -side cluster with any B -side cluster.

Combining (i) and (ii): the gauge quotient $Q_n(G_A \sqcup G_B, \psi)$ has cardinality $|Q_n(G_A, \psi_A)| + |Q_n(G_B, \psi_B)|$, i.e. $\delta_\sqcup = 0$. $\square$

The bridge ($\delta_{\bowtie_k} = +3k$) and glue ($\delta_{\#_g} = -(4g - 3)$) cases require finer accounting of which specific clusters are created or merged at the interface. We have not produced closed-form structural proofs of these. The empirical evidence (8 (A, B) pairs, 6 join modes, exact agreement across all non-degenerate cases) is strong but does not constitute a proof in the spec-status sense; the closed-form structural proofs are flagged as open in §11.3.

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